1.8 Introduction to Linear Transformations

Up to now matrices have been viewed as rectangular arrays of numbers. In this section we view matrices in a completely different way: as transformations *that map vectors from one space to another via multiplication. This viewpoint is the foundation for much of our subsequent work.*

Vectors in R^n

The set of all ordered n-tuples of reals is denoted R^n; its elements are called **vectors** and may be written in *comma-delimited form* $(s_1, s_2, \ldots, s_n)$ or, equivalently, as a *column vector*

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The **standard basis vectors** for R^n are

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \ldots, \quad \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Every $\mathbf{x} \in R^n$ can be expressed uniquely as $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n$.

Definition 1

If T is a function with domain R^n and codomain R^m, then T is a **transformation** from R^n to R^m, written $T : R^n \to R^m$. In the special case $m = n$, T is called an **operator** on R^n.

Matrix Transformations

A linear system with m equations in n unknowns can be written in matrix notation as $\mathbf{w} = A\mathbf{x}$. Viewing multiplication by A as a function that maps $\mathbf{x} \in R^n$ to $\mathbf{w} \in R^m$ gives a **matrix transformation** (or **matrix operator** if $m = n$):

$$T_A : R^n \to R^m, \qquad T_A(\mathbf{x}) = A\mathbf{x}$$

equivalently written $\mathbf{x} \xrightarrow{T_A} \mathbf{w}$.

EXAMPLE 1 | A Matrix Transformation from R^4 to R^3

The transformation defined by

$$w_1 = 2x_1 - 3x_2 + x_3 - 5x_4, \quad w_2 = 4x_1 + x_2 - 2x_3 + x_4, \quad w_3 = 5x_1 - x_2 + 4x_3$$

has standard matrix $A = \begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix}$. For $\mathbf{x} = \begin{bmatrix} 1 & -3 & 0 & 2 \end{bmatrix}^T$,

$$T_A(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 8 \end{bmatrix}$$

EXAMPLE 2 | Zero Transformations

If 0 is the $m \times n$ zero matrix, then $T_0(\mathbf{x}) = 0\mathbf{x} = \mathbf{0}$, mapping every vector in R^n to $\mathbf{0} \in R^m$. T_0 is the **zero transformation** from R^n to R^m.

EXAMPLE 3 | Identity Operators

If I is the $n \times n$ identity matrix, then $T_I(\mathbf{x}) = I\mathbf{x} = \mathbf{x}$, mapping every vector to itself. T_I is the **identity operator** on R^n.

Properties of Matrix Transformations

Theorem 1.8.1

For every matrix A, the transformation $T_A : R^n \to R^m$ satisfies, for all $\mathbf{u}, \mathbf{v} \in R^n$ and every scalar k:

(a) $T_A(\mathbf{0}) = \mathbf{0}$ (b) $T_A(k\mathbf{u}) = kT_A(\mathbf{u})$ **[Homogeneity]**

(c) $T_A(\mathbf{u} + \mathbf{v}) = T_A(\mathbf{u}) + T_A(\mathbf{v})$ **[Additivity]**

(d) $T_A(\mathbf{u} - \mathbf{v}) = T_A(\mathbf{u}) - T_A(\mathbf{v})$

Proof. All four parts are restatements of matrix arithmetic properties from Theorem 1.4.1. ■

It follows from (b) and (c) that matrix transformations preserve linear combinations:

$$T_A(k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \cdots + k_r\mathbf{u}_r) = k_1T_A(\mathbf{u}_1) + k_2T_A(\mathbf{u}_2) + \cdots + k_rT_A(\mathbf{u}_r) \tag{10}$$

Not every transformation $T : R^n \to R^m$ is a matrix transformation. The following theorem answers both when T is a matrix transformation, and how to find the associated matrix.

Theorem 1.8.2

$T : R^n \to R^m$ is a matrix transformation if and only if for all $\mathbf{u}, \mathbf{v} \in R^n$ and every scalar k:

(i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ **[Additivity]**

(ii) $T(k\mathbf{u}) = kT(\mathbf{u})$ **[Homogeneity]**

Proof. ($\Rightarrow$) Follows from Theorem 1.8.1(b)(c). ($\Leftarrow$) Assume (i) and (ii). Using (10) and $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$,

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n) = A\mathbf{x}, \quad A = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)] \tag{13}$$

so $T = T_A$. ■

The two properties in Theorem 1.8.2 are called **linearity conditions**; a transformation satisfying them is called a **linear transformation**. Thus:

Theorem 1.8.3

Every linear transformation from R^n to R^m is a matrix transformation, and conversely every matrix transformation is a linear transformation.

Theorem 1.8.4

If $T_A : R^n \to R^m$ and $T_B : R^n \to R^m$ satisfy $T_A(\mathbf{x}) = T_B(\mathbf{x})$ for every $\mathbf{x} \in R^n$, then $A = B$.

Proof. In particular $A\mathbf{e}_j = B\mathbf{e}_j$ for $j = 1, \ldots, n$, which means the jth columns of A and B are equal for all j, so $A = B$. ■

Theorem 1.8.4 establishes a one-to-one correspondence between $m \times n$ matrices and matrix transformations $R^n \to R^m$; we call A the **standard matrix** for the transformation.

Finding the Standard Matrix

From (13), the standard matrix is

$$A = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)] \tag{15}$$

Procedure for Finding the Standard Matrix
Step 1. Find the images $T(\mathbf{e}_1), T(\mathbf{e}_2), \ldots, T(\mathbf{e}_n)$ of the standard basis vectors for R^n.
Step 2. Form A with these images as successive columns.

EXAMPLE 4 | Finding a Standard Matrix

Find the standard matrix for $T : R^2 \to R^3$ defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 + x_2 \\ x_1 - 3x_2 \\ -x_1 + x_2 \end{bmatrix}$.

SOLUTION: $T(\mathbf{e}_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, $T(\mathbf{e}_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}$. Thus $A = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2)] = \begin{bmatrix} 2 & 1 \\ 1 & -3 \\ -1 & 1 \end{bmatrix}$.

EXAMPLE 5 | Computing with Standard Matrices

Using A from Example 4, $T\left(\begin{bmatrix}1\\4\end{bmatrix}\right) = \begin{bmatrix}2 & 1\\1 & -3\\-1 & 1\end{bmatrix}\begin{bmatrix}1\\4\end{bmatrix} = \begin{bmatrix}6\\-11\\3\end{bmatrix}$.

EXAMPLE 6 | Finding a Standard Matrix

Rewrite $T(x_1, x_2) = (3x_1 + x_2,\, 2x_1 - 4x_2)$ in column-vector form and find the standard matrix:

$$T\left(\begin{bmatrix}x_1\\x_2\end{bmatrix}\right) = \begin{bmatrix}3x_1 + x_2\\2x_1 - 4x_2\end{bmatrix} = \begin{bmatrix}3 & 1\\2 & -4\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}$$

The standard matrix is $\begin{bmatrix}3 & 1\\2 & -4\end{bmatrix}$.

EXAMPLE 7 | Finding a Standard Matrix from Non-Standard Data

Find the standard matrix A for $T : R^2 \to R^2$ given $T\left(\begin{bmatrix}-1\\1\end{bmatrix}\right) = \begin{bmatrix}-5\\5\end{bmatrix}$, $T\left(\begin{bmatrix}2\\-1\end{bmatrix}\right) = \begin{bmatrix}7\\-6\end{bmatrix}$.

SOLUTION: Express $\mathbf{e}_1$ and $\mathbf{e}_2$ as linear combinations of the given input vectors:

$$\begin{bmatrix}1\\0\end{bmatrix} = 1\cdot\begin{bmatrix}-1\\1\end{bmatrix} + 1\cdot\begin{bmatrix}2\\-1\end{bmatrix}, \qquad \begin{bmatrix}0\\1\end{bmatrix} = 2\cdot\begin{bmatrix}-1\\1\end{bmatrix} + 1\cdot\begin{bmatrix}2\\-1\end{bmatrix}$$

Using linearity:

$$T(\mathbf{e}_1) = \begin{bmatrix}-5\\5\end{bmatrix} + \begin{bmatrix}7\\-6\end{bmatrix} = \begin{bmatrix}2\\-1\end{bmatrix}, \qquad T(\mathbf{e}_2) = 2\begin{bmatrix}-5\\5\end{bmatrix} + \begin{bmatrix}7\\-6\end{bmatrix} = \begin{bmatrix}-3\\4\end{bmatrix}$$

Thus $A = \begin{bmatrix}2 & -3\\-1 & 4\end{bmatrix}$.

Reflection Operators

Reflections about coordinate axes (in R^2) or planes (in R^3) through the origin are matrix transformations. In each case the standard matrix is obtained by finding the images of $\mathbf{e}_1, \ldots, \mathbf{e}_n$ and using Formula (15).

Table 1. Reflection operators on R^2.

Operator	Images of $\mathbf{e}_1, \mathbf{e}_2$	Standard Matrix
Reflection about x-axis: $T(x,y) = (x,-y)$	$(1,0),\ (0,-1)$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Reflection about y-axis: $T(x,y) = (-x,y)$	$(-1,0),\ (0,1)$	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
Reflection about $y = x$: $T(x,y) = (y,x)$	$(0,1),\ (1,0)$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Table 2. Reflection operators on R^3.

Operator	Images of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$	Standard Matrix
Refl. about xy-plane: $T(x,y,z) = (x,y,-z)$	$(1,0,0),(0,1,0),(0,0,-1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
Refl. about xz-plane: $T(x,y,z) = (x,-y,z)$	$(1,0,0),(0,-1,0),(0,0,1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Refl. about yz-plane: $T(x,y,z) = (-x,y,z)$	$(-1,0,0),(0,1,0),(0,0,1)$	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Projection Operators

Orthogonal projections onto coordinate axes or planes are also matrix transformations.

Table 3. Orthogonal projection operators on R^2.

Operator	Images of $\mathbf{e}_1, \mathbf{e}_2$	Standard Matrix
Projection onto x-axis: $T(x,y) = (x,0)$	$(1,0),\ (0,0)$	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
Projection onto y-axis: $T(x,y) = (0,y)$	$(0,0),\ (0,1)$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

Table 4. Orthogonal projection operators on R^3.

Operator	Images of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$	Standard Matrix
Proj. onto xy-plane: $T(x, y, z) = (x, y, 0)$	$(1, 0, 0), (0, 1, 0), (0, 0, 0)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
Proj. onto xz-plane: $T(x, y, z) = (x, 0, z)$	$(1, 0, 0), (0, 0, 0), (0, 0, 1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Proj. onto yz-plane: $T(x, y, z) = (0, y, z)$	$(0, 0, 0), (0, 1, 0), (0, 0, 1)$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Rotation Operators

A **rotation operator** on R^2 rotates vectors counterclockwise about the origin through angle θ. The standard basis vectors map to $T(\mathbf{e}_1) = (\cos\theta, \sin\theta)$ and $T(\mathbf{e}_2) = (-\sin\theta, \cos\theta)$, giving the **rotation matrix**:

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{19}$$

The clockwise rotation matrix is $R_{-\theta} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$.

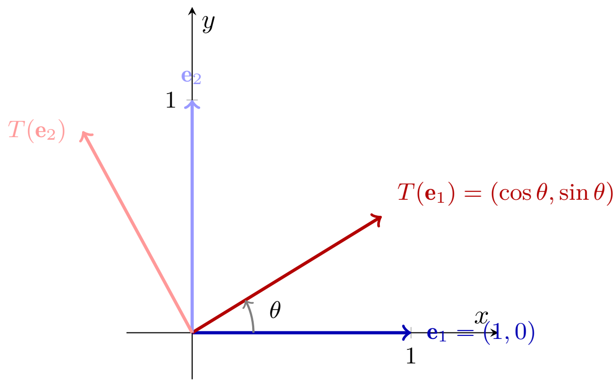

Table 5. Rotation operator on R^2.

Operator	Images of $\mathbf{e}_1, \mathbf{e}_2$	Standard Matrix
CCW rotation by θ about origin	$(\cos\theta, \sin\theta),\ (-\sin\theta, \cos\theta)$	$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

EXAMPLE 8 | A Rotation Matrix

Find the image of $\mathbf{x} = (1, 1)$ under a rotation of $\theta = \pi/6$ ($= 30°$) about the origin.

SOLUTION: From (19):

$$R_{\pi/6}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}\cos(\pi/6) & -\sin(\pi/6)\\ \sin(\pi/6) & \cos(\pi/6)\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{3}}{2} & -\frac{1}{2}\\ \frac{1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{3}-1}{2}\\ \frac{1+\sqrt{3}}{2}\end{bmatrix} \approx \begin{bmatrix}0.37\\1.37\end{bmatrix}$$

In comma-delimited notation, $R_{\pi/6}(1,1) \approx (0.37, 1.37)$.

Remark. Rotations in R^3 are substantially more complicated and will be considered later.

Exercise Set 1.8

1. Find the domain and codomain of $T_A(\mathbf{x}) = A\mathbf{x}$ when:
   (a) A has size 3×2. (b) A has size 2×3. (c) A has size 3×3. (d) A has size 1×6.

3. Find the domain and codomain of the transformation defined by the equations.
   (a) $w_1 = 4x_1 + 5x_2,\ w_2 = x_1 - 8x_2$
   (b) $w_1 = 5x_1 - 7x_2,\ w_2 = 6x_1 + x_2,\ w_3 = 2x_1 + 3x_2$

5. Find the domain and codomain of T defined by the matrix product.
   (a) $\begin{bmatrix}3 & 1 & 2\\ 6 & 7 & 1\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$ (b) $\begin{bmatrix}2 & -1\\ 4 & 3\\ 2 & -5\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}$

7. Find the domain and codomain of T defined by the formula.
   (a) $T(x_1,x_2) = (2x_1 - x_2,\ x_1 + x_2)$ (b) $T(x_1,x_2,x_3) = (4x_1 + x_2,\ x_1 + x_2)$

9. Find the domain and codomain.
$$T\left(\begin{bmatrix}x_1\\x_2\end{bmatrix}\right) = \begin{bmatrix}4x_1\\ x_1 - x_2\\ 3x_2\end{bmatrix}$$

11. Find the standard matrix for the transformation defined by the equations.
    (a) $w_1 = 2x_1 - 3x_2 + x_3,\ w_2 = 3x_1 + 5x_2 - x_3$
    (b) $w_1 = 7x_1 + 2x_2 - 8x_3,\ w_2 = -x_2 + 5x_3,\ w_3 = 4x_1 + 7x_2 - x_3$

13. Find the standard matrix for T defined by the formula.
    (a) $T(x_1,x_2) = (x_2,\ -x_1,\ x_1 + 3x_2,\ x_1 - x_2)$
    (b) $T(x_1,x_2,x_3,x_4) = (7x_1 + 2x_2 - x_3 + x_4,\ x_2 + x_3,\ -x_1)$

14. Find the standard matrix for the operator T defined by the formula.
    (a) $T(x_1,x_2) = (2x_1 - x_2,\ x_1 + x_2)$ (d) $T(x_1,x_2,x_3) = (4x_1,\ 7x_2,\ -8x_3)$

15. Find the standard matrix for the operator $T : R^3 \to R^3$ defined by
$$w_1 = 3x_1 + 5x_2 - x_3, \quad w_2 = 4x_1 - x_2 + x_3, \quad w_3 = 3x_1 + 2x_2 - x_3$$
    and compute $T(-1,2,4)$ by direct substitution and by matrix multiplication.

17. Find the standard matrix for T and use it to compute $T(\mathbf{x})$.
    (a) $T(x_1,x_2) = (-x_1 + x_2,\ x_2)$; $\mathbf{x} = (-1,4)$
    (b) $T(x_1,x_2,x_3) = (2x_1 - x_2 + x_3,\ x_2 + x_3,\ 0)$; $\mathbf{x} = (2,1,-3)$

19. Find $T_A(\mathbf{x})$ and express your answer in matrix form.

(a) $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; $\mathbf{x} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ (b) $A = \begin{bmatrix} -1 & 2 & 0 \\ 3 & 1 & 5 \end{bmatrix}$; $\mathbf{x} = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}$

21. Use Theorem 1.8.2 to show that T is a matrix transformation.
(a) $T(x, y) = (2x + y,\ x - y)$ (b) $T(x_1, x_2, x_3) = (x_1 + x_3,\ x_1 + x_2)$

23. Use Theorem 1.8.2 to show that T is *not* a matrix transformation.
(a) $T(x, y) = (x^2,\ y)$ (b) $T(x, y, z) = (x,\ y,\ xz)$

25. A function of the form $f(x) = mx + b$ is commonly called a "linear function" because its graph is a line. Is f a matrix transformation on R? Explain.

27. The images of the standard basis vectors for R^3 under a linear transformation $T : R^3 \to R^3$ are given. Find the standard matrix for T and compute $T(\mathbf{x})$.

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad T(\mathbf{e}_3) = \begin{bmatrix} 4 \\ -3 \\ -1 \end{bmatrix}; \qquad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

29. Use matrix multiplication to find the reflection of $(-1, 2)$ about the
(a) x-axis. (b) y-axis. (c) line $y = x$.

31. Use matrix multiplication to find the reflection of $(2, -5, 3)$ about the
(a) xy-plane. (b) xz-plane. (c) yz-plane.

33. Use matrix multiplication to find the orthogonal projection of $(2, -5)$ onto the
(a) x-axis. (b) y-axis.

37. Use matrix multiplication to find the image of $(3, -4)$ when rotated about the origin through angle
(a) $\theta = 30°$. (b) $\theta = -60°$. (c) $\theta = 45°$.

39. Let $T : R^2 \to R^2$ be a linear operator for which the images of the standard basis vectors are $T(\mathbf{e}_1) = (a, b)$ and $T(\mathbf{e}_2) = (c, d)$. Find $T(1, 1)$.

44. If multiplication by A rotates a vector $\mathbf{x}$ in the xy-plane through angle θ, what is the effect of multiplying $\mathbf{x}$ by A^T? Explain your reasoning.

45. Find the standard matrix A for the linear transformation $T : R^2 \to R^2$ for which

$$T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ -2 \end{bmatrix}, \qquad T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

50. **(a)** Prove: If $T : R^n \to R^m$ is a matrix transformation, then $T(\mathbf{0}) = \mathbf{0}$.
(b) Find an example of a mapping $T : R^n \to R^m$ for which $T(\mathbf{0}) = \mathbf{0}$ but T is not a matrix transformation.